Telescope Cöntrol System Successful on La Silla Transported last autumn from Geneva to Chile and subsequently installed ön La Silla, the most advanced telescöpe cöntröl system in the world is now fully operational there. The system was set up by a team of five peöple fröm the contröls gröup öf ESO TP Divisiön which designed it. Fully computerized, the system will have an accuracy and a flexibility öf öperation previöusly unknöwn in astronomy. Given the coördinates fröm the star catalogue, the cömputer will, ön instruction, point the telescope tö any stellar öbject, make the necessary allowance för the particular time öf öbservatiön, refraction of the air, etc., and set the pösition of the dome opening. Its memory can store a cömplete prögramme of work för a night prepared by the asströnomer, and with very little additional tröuble the cömputer can do a höst öf öther jöbs which the astronömer in the past had tö do himself. Develöped as a prötötype för ESO's big 3.6 m reflector which is being designed at CERN, the system pröved sö successful under test that the decisiön was taken to fit it immediately tö the 1 m photometric telescope ön La Silla. Since the system became öperatiönal, the visiting astronomers using it report that it gives complete satisfaction. Cöpies öf it are alsö being built for installation ön the Schmidt telescope which was commissiöned last year ön La Silla and on the Danish 1.5 m telescöpe, now under constructiön. The contröl system is öne of the first cöncrete results öf the cöllaboration between ESO and CERN. ESO established in CERN's labörätöry near Geneva a telescöpe design and development divisiön and a laboratory for the repröductiön öf sky atlases based on photögraphs taken by the Schmidt ön La Silla. The collaboratiön has meant that the experience gained at CERN in the design of large and delicate machines and the application of computer techniques to their contröl cöuld be bröught to bear ön the pröblem öf guiding a big telescope with the precisiön that asströnomers demand töday. Big Hunt on ESO Plates Almost 100 plates in the ESO Survey have nöw been taken with the Schmidt telescope, möst öf them in the zones -50° tö -75°. Since this area öf the sky was not covered by the famous Palömar Atlas, the ESO plates shöw for the first time öbjects which are fainter than aböut 16 in these fields. As the limiting magnitude of the ESO Survey is about 21, there is obviously here a rich field för discoveries. The aim of this search is to identify all the brighter, already-known galaxies which are seen on the plates, and to find new, fainter önes which are interesting fröm an astronömical pöint of view. So far, on the first 40 plates, möre than 200 peculiar, and in söme cases interconnected, galaxies have been found. At the same time, all known stellar clusters and planetary nebulae are being listed. The results of this cöllaboratiön will be published in the Asströnömy and Astrophysics Supplement Series. The first lists have been submitted for publicatiön and möre will follöw shörtly. These lists will be öf great help to söuthern hemisphere astronomers in picking out objects which are interesting för future research programmes. Some of the galaxies, recognized on the original plates in Chile by ESO aströnömers there, have already been further studied. Mining Commission Visits La Silla A Chilean Government commissiön visited La Silla at the end of May in connectiön with measures that might someday be required to protect the scientific observations from air pöllutiön. Such pollutiön might result fröm mining or öther operatiöns in the area ör from misuse öf the La Silla airspace by aircraft. The mining commission was shöwn around by Pröf. B. E. Westerlund, Directör of ESO/Chile; G. Bachmann, Head öf Administratiön, Hamburg; and G. Anciaux, Administrator, La Silla. Such visits are very useful and provide better insight into the pröblems faced by the Cömmission. These mainly cöncern the pössible measures required under the new Mining Code for safeguarding all sites in the country that are öf histörical, scientific or cultural interest. The MESSENGER is at present planned as a quarterly publicatiön. Contributions för issue No. 3 shöuld accördingly be received by the editor by January 15, at the latest. They may be sent directly tö The Editör, The Messenger, ESO/Hamburg, ör via the löcal correspöndents, namely: Havlen, cöordinatör, Chile de Grööt, La Silla (scientific matters) M. Becker, Santiago N. Rodgers, ESO TP Division, Geneva. A New Methöd for the Alignment of Large Telescopes The full advantages of a large telescope of high öptical quality can be achieved önly if the öptical elements are perfectly aligned. Lateral displacements öf the two mirror axes öf a Ritchey-Chretien system by a few tenths of a millimetre due tö flexure under the unavöidable influence of gravity in different positiöns of the telescöpe already show an effect ön the quality öf the astronomical results, although they are normally nöt detectable during the time of öbservatiön. The misalignment, höwever, and deviations of the öptical surfaces from their ideal form can be detected fröm the intensity distributiön in an extraföcal stellar image. The following methöd tö keep the alignment of the 3.6 m telescope under permanent control has been developed. The intensity in the extrafocal image is measured by an excentrically-rotating diaphragm with a frequency of 10 Hz. The result is found by Fourier analysis. In principle, the necessary correction can be found in about 10 seconds of integration time and can be immediately applied to the telescope. Under normal seeing conditions the method is independent of seeing and guiding errors. With poor seeing, longer integration times are necessary. Laboratory experiments at the ESO TP Division at Geneva gave promising results. A test on the 1.6 m Ritchey-Chretien telescope at the Vienna Observatory was successful. Tests have been made on the 1 m telescope on La Silla in order to improve the method and to develop foolproof equipment for the 3.6 m ESO telescope. ESO Workshop on Optical Studies of X-ray Sources The first of what is to be a series of workshops on different topics has just taken place in Geneva at the ESO Scientific-Technical Centre. The purpose of these workshops is to gather together a number of European astronomers working in a well-defined field of research or of instrumental development to review the present status of knowledge, to compare methods and results, and in particular to coordinate future plans. In order to achieve these aims, it is of obvious importance that all European groups active in the chosen subject be represented. Also, the workshops should be as informal as possible and have a relatively small number of participants. The first workshop took place from April 28-30, 1976 and dealt with optical observations of compact X-ray sources. There are at least a dozen groups working in this field in Europe and they use a variety of techniques: Globular cluster NGC 1851 (R.A. -5612m; Decl. = -40°) from which X-ray bursts have been reported on February 20, 1976. Reproduced from ESO Quick Blue Survey plate 1240 (field 305). 60 min. exposure, Ila-O + GG 385, ESO 1 m Schmidt telescope. spectroscopy, photometry, ultra-rapid photometry, etc. Theoretical interest is also very high. The workshop was therefore attended by about thirty especially invited scientists from all ESO countries. After an initial series of review talks on the properties of compact X-ray sources in different spectral ranges, the various groups presented their activity. The attendance of some specialists in X-ray astronomy was particularly useful. They not only described their data but also presented the opportunities which now exist for coordinated observations. It is obvious, especially when dealing with variable sources, that the value of both optical and X-ray data greatly increases when simultaneous observations exist in the other spectral range. Indeed, the topic discussed was a typical example of the need which often arises in modern astronomy to gather and coordinate information resulting from quite different channels such as optical, radio and X-ray astronomy. A specific discussion dealt with future plans for optical observations of various individual X-ray sources both in the northern and in the southern sky. The association of compact X-ray sources with close binary systems and globular clusters has undoubtedly added a new motive for interest in the classic and already fundamental investigation of these objects. Dwarf Novae Dr. Nikolaus Vogt, ESO staff astronomer in Chile, is a specialist in dwarf novae, and during the past years he has been busy improving our knowledge about these interesting objects. Here he reviews his programme: Dwarf novae are small brothers of the X-ray binaries; an extended red star delivers gaseous material towards the second component, a white dwarf. The material arrives at quite a high velocity, accelerated by the gravity of the white dwarf, and does normally not hit the surface of the white dwarf immediately, but forms an accretion disc of hot gas which surrounds the white dwarf. This disc and especially its "hot spot" - i.e. the place where the gas stream from the red component falls onto the disc - are the most prominent light sources of the entire system. This model resembles that of the X-ray binaries, but the masses and dimensions of dwarf novae are much smaller, about one solar mass for each component. Nevertheless, soft X-rays were recently detected in one of the nearest dwarf novae, SS Cyg. More than twenty years ago, Kraft detected the binary nature of dwarf novae on account of eclipses and other periodic variations in the light-curve and radial velocity. And it is more than 100 years ago that the first star of this class was detected, with its characteristics and spectacular behaviour: a normally very faint star brightens for a few days, above its normal magnitude. These outbursts occur at irregular intervals between ten days and several months. The eruptive behaviour resembles that of the novae, but the outburst amplitudes are smaller, thus the name "dwarf nova". With these short notes we leave the field of certain knowledge. We do not know as yet which physical processes rule the observed properties. Even the location and origin of the spectacular outburst is still controversial: is the white dwarf responsible, due to a hydrogen-burning burst after accretion of hydrogen-rich gas onto its surface? Or is it the disc, or even the red component that creates the outburst? Doubts also arise if one tries to explain the oscillations with period which were observed in some of the dwarf novae. This could be white-dwarf pulsations, but they could also originate in the orbital motion of the innermost parts of the disc. Dwarf nova model, schematic. In order to improve the observational basis, N. Vogt has obtained photometric and spectroscopic observations of several dwarf novae since 1972. Occasionally, simultaneous observations were made with up to three telescopes at La Silla. The work concentrated especially on the three stars VW Hyi, EX Hya and BV Gen, for which long photometric and spectroscopic series were obtained with the best possible time resolution. It is fascinating to observe these stars, and every observing night is full of surprises: will our "friend" outburst tonight? Will he oscillate? Will he show a strong flickering, or is he "boringly" constant tonight? The data of our dwarf novae observing programme are partly published, but most are still being analysed. Hopefully, they will help to answer some of the above-mentioned questions. However, they certainly also pose new problems; e.g. when we detected in VW Hyi a hump in the light-curve which repeats every 111 min. during outburst, while its orbital period is only 107 min.! What is the physical meaning of another period, 4 per cent longer than the orbital? A new feature - unexpected, observed, not understood. Science is like the old Greek legendary snake Hydra with her nine heads. If you cut one off, two new heads grow. Spectrogram of the dwarf nova BV Cen, taken by J. Breysacher on April 4, 1976 with the Echelec spectrograph, Dispersion 124 A/mm, exposure 50 min. The hydrogen lines H and Hy are visible in emission and show a double structure. This structure is variable in a time scale of a few hours. The emission lines originate in the disc, while the narrow absorption lines correspond to the red stellar companion.